G. NARAYANAMMA INSTITUTE OF TECHNOLOGY AND SCIENCE

(AUTONOMOUS) (For Women)

Shaikpet, Hyderabad - 500 104

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

IV B.Tech I Semester Academic Year 2026-27
Case Study

SUBJECT: DATA SCIENCE USING R
SUBJECT CODE: 127EY **Max Marks: 5**
BRANCH: CSE-A, B, C and D

Students are required to perform **one** complete data analysis and predictive modeling workflow in R using the datasets specified below. The analysis should include data exploration, data quality assessment, visualization, cleaning, sampling, model building, evaluation, prediction, and business interpretation.

CASE STUDY 1 – E-Commerce Customer and Sales Analysis

Dataset: UCI Online Retail II Dataset

The **Online Retail II** dataset contains more than one million transaction records from a UK-based online retailer covering a period of two years. The dataset contains information such as invoice number, product description, quantity, invoice date, unit price, customer ID, and country. The dataset also contains missing values and other data-quality issues that need to be identified and handled appropriately.

Objective:
Analyze customer purchasing and sales patterns and develop predictive models to derive meaningful business insights from the transaction data.

Dataset:
UCI Online Retail II Dataset (https://archive.ics.uci.edu/dataset/502/online+retail+ii)

CASE STUDY 2 – Income Classification Using Census Data

The dataset contains census information about individuals, including age, education, occupation, marital status, working class, hours worked per week, and other demographic and employment-related attributes.

The objective is to build a classification model to **predict whether an individual's annual income exceeds $50,000 based on census data**.

Objective:
Explore the demographic and employment characteristics of individuals and develop predictive models for income classification.

Dataset:
UCI Dataset (https://archive.ics.uci.edu/dataset/2/adult)

CASE STUDY 3 – Telecom Customer Churn Prediction

Dataset: KDD Cup 2009

The **KDD Cup 2009** dataset was developed for customer relationship management (CRM) and contains telecom-related customer information. The competition involved predicting **churn, appetency, and up-selling**.

The dataset contains both numerical and categorical variables and includes missing values. The small version contains approximately 230 variables, while the large version contains approximately 15,000 variables.

Objective:
Analyze customer characteristics and develop predictive models to identify customers who are likely to churn.

Dataset:
KDD Cup 2009 https://www.kdd.org/kdd-cup/view/kdd-cup-2009/Data (Small version 230 var.)

CASE STUDY 4 – House Price Prediction Using Machine Learning

Dataset: House Prices – Advanced Regression Techniques
https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques

Problem Description:
The dataset contains information about residential properties in Ames, Iowa, including property size, location, construction year, quality, and other characteristics. The objective is to predict the sale price of a house based on these features. The reference notebook demonstrates the use of TensorFlow Decision Forests for house price prediction.

Objective:
Perform exploratory data analysis, identify and handle data-quality issues, and develop regression models to predict house prices.

CASE STUDY 5 - Airline Passenger Satisfaction Prediction

Dataset/Competition: Kaggle Playground Series – Season 6, Episode 10
Link: https://www.kaggle.com/competitions/playground-series-s6e10

Problem Description:
Airline passenger satisfaction is an important factor influencing customer loyalty, repeat bookings, and an airline's business performance. Passenger satisfaction may depend on several factors, including flight distance, seat comfort, inflight services, departure and arrival delays, and the quality of airport services.

In this case study, students will analyze airline passenger data and develop a machine learning model to predict whether a passenger is satisfied with the airline experience.

Objective:
To perform exploratory data analysis, identify factors associated with passenger satisfaction, and develop and evaluate classification models for predicting airline passenger satisfaction

Case Study 6: Retail Sales Forecasting

Problem: Retailers need to predict future sales to manage inventory, plan promotions, and avoid overstocking or stock shortages.

Dataset: Kaggle – Store Sales: Time Series Forecasting

Case study 7: Tourism Demand Forecasting

Problem: Tourism authorities and hospitality businesses need to anticipate visitor arrivals to plan accommodation, transport, and staffing.

Dataset: World Bank – International Tourism Data

The case study report should have these sections 1. Introduction (Business problem, Objective), 2. Dataset Description, 3. Data Loading, 4. Data Exploration, 5. Data Quality, 6. Data Visualization, 7. Data Cleaning, 8. Sampling and Validation, 9. Modeling (Single-variable model, Multi-variable model) 10. Model Evaluation, 11. Predictions, 12. Business Interpretation, 13. Conclusion

Rubrics for evaluation

Criteria	Description	Marks
Case Study Submission	Timely and appropriate submission of the case study with all required components	2 marks
Explanation & Presentation	Clear explanation of the approach, analysis, and results during discussion or viva	2 marks
Revisions & Improvements	Implementation of suggested modifications or improvements after feedback	1 marks

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